Cloud Architecture

Centralised, decentralised and distributed structures

Summary Notes

Contents

Lesson objectives

By the end of this lesson, you should be able to:

- Identify the various IT structures
- Describe the differences between each structure
- Review the architectural structures

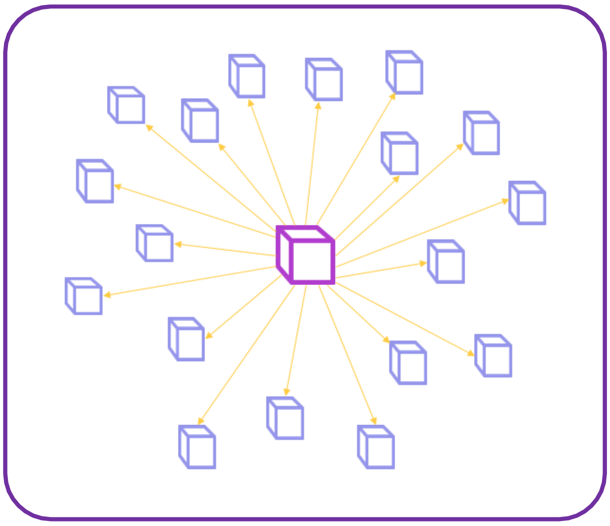
IT structures

Choosing the correct IT structure for an organisation requires careful thought. The choice of IT structure depends on the specific needs and constraints of the organisation, and it is important for organisations to consider these factors when making a decision. There are several types of IT structures that organisations can use to manage their information technology resources, including:

- Centralised
- Decentralised
- Distributed

Centralised structure

A centralised IT structure involves having all IT resources, such as hardware, software, and personnel, controlled and managed by a central IT department. This structure is best suited for organisations with a small IT staff and a limited number of IT resources. The client/server architecture of these structures are usually located in a single physical location. For example, intranets are usually highly centralised.



Characteristics of centralised structures

- Single global clock
- Single central node coordinates all nodes
- All nodes dependant on the operation of central node

Advantages of centralised structures

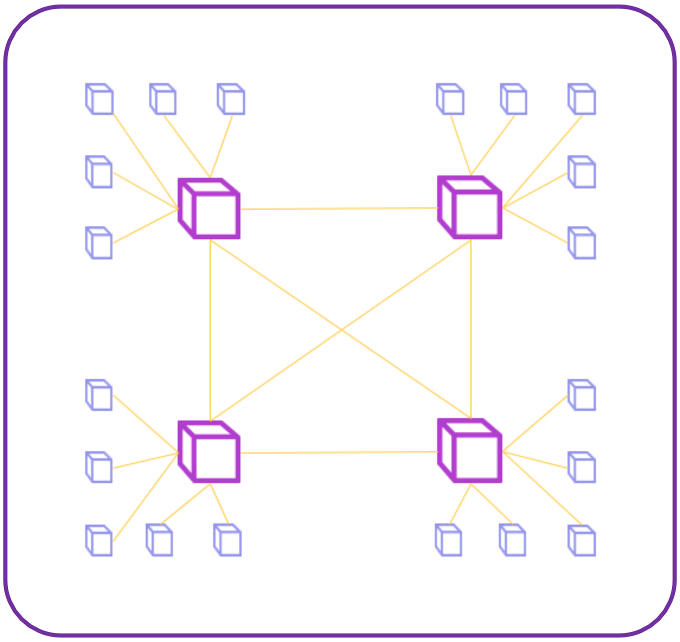
- Easy physical implementation
- Dedicated computational resources
- High performance
- More cost-efficient for small systems

Disadvantages of centralised structures

- Dependent on network stability
- Dependent on central node
- Lack of data backup
- Server updates may cause downtime

Decentralised structure

A decentralised IT structure involves having IT resources and decision-making spread throughout the organisation, rather than being centralised in a single IT department. Each node on the network is independent and the system behaviour is dictated by the aggregate decision of each node. Bitcoin is regarded as the most decentralised network.



Characteristics of decentralised structures

- Lack of global clock
- Multiple central nodes
- Failure of a single node does not cause system failure

Advantages of decentralised structures

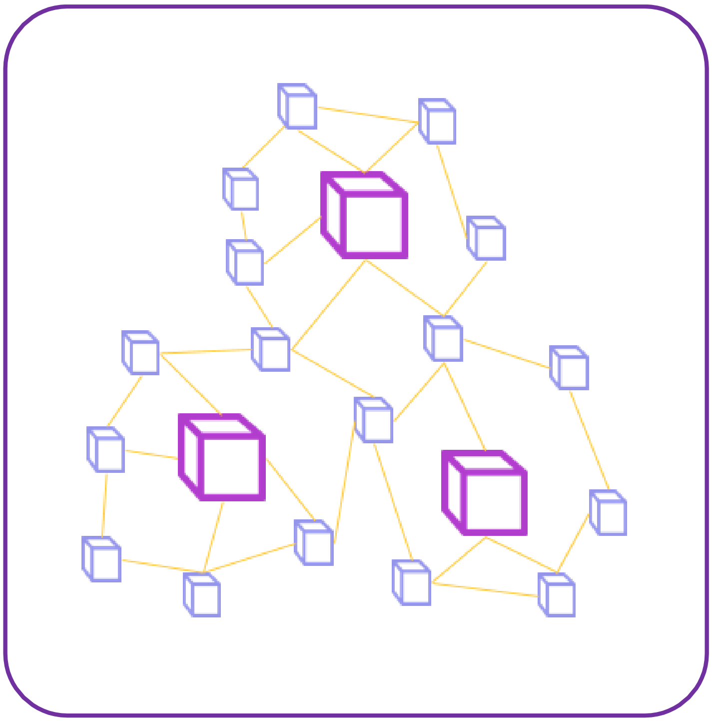
- Less performance bottlenecks
- High system availability
- More autonomy over resources

Disadvantages of decentralised structures

- No obvious chain of command
- Lack of regulatory oversight
- Difficulty determining single node failure

Distributed structure

A distributed system is a collection of computer programs that utilise computational resources across multiple, separate computation nodes to achieve a common, shared goal. Also known as distributed computing or distributed databases, it relies on separate nodes to communicate and synchronise over a common network. These nodes typically represent separate physical hardware devices but can also represent separate software processes, or other recursive encapsulated systems.



Characteristics of distributed structures

- Shared resources
- Simultaneous processing
- Scalable structure

Advantages of distributed structures

- High level of transparency
- Very low latency
- Easier to back up data

- System availability

Disadvantages of distributed structures

- More prone to overloading
- Expensive to implement on a large scale
- Software can be very complex